

Amazon Product Dataset Visualization

Tools Used: Python, Pandas, Seaborn, Matplotlib

Objective of Visualization

The objective of this task is to transform Amazon product data into meaningful visualizations that:

- Reveal hidden patterns

- Improve business understanding

- Support data-driven decision-making

This report focuses purely on visual insights, not data cleaning.

Dataset Overview

- Total Products:1,400+
- Domain:E-commerce(Amazonproduct data)
- Key attributes used for visualization: Category–
Product classification Actual Price &
Discounted Price–Pricing analysis Discount
Percentage–Promotion insights Rating–
Customer satisfaction
- ReviewCount–Product popularity
- This dataset represents real-world e-commerce data and is well-
- suitedforvisualanalytics andbusinessdecision-making.

Visualization 1: Top 10 Product Category Distribution

Chart Type: Horizontal Bar Chart

Tool: Seaborn

6-

What it shows:

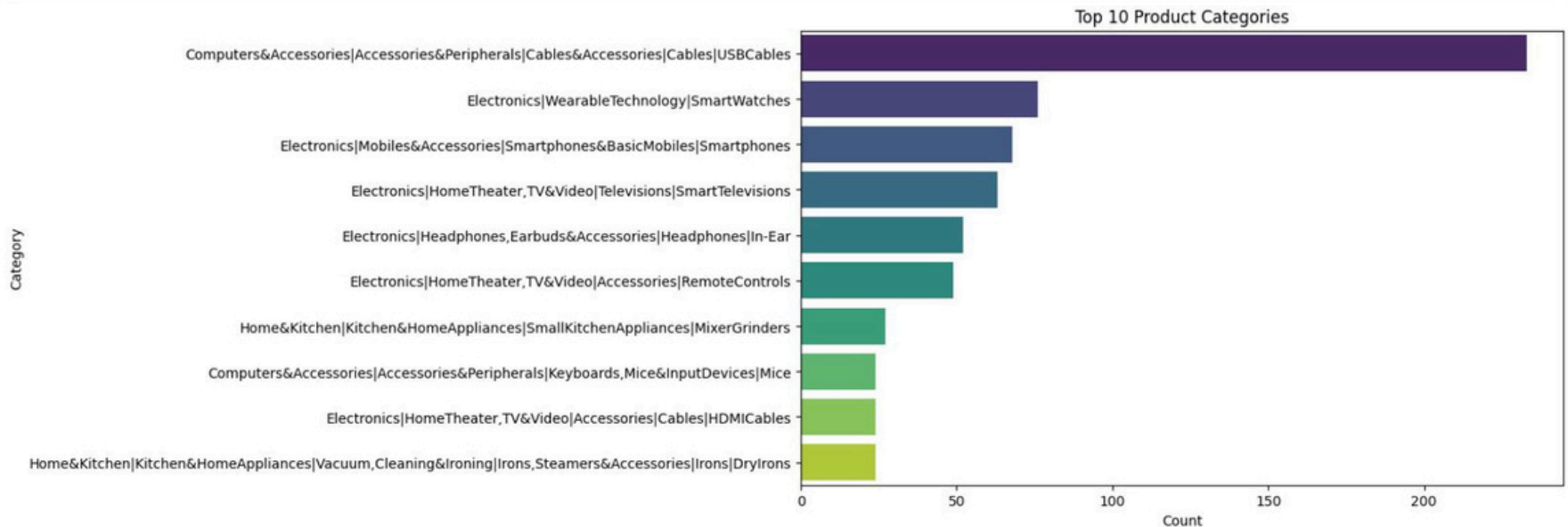
Distribution of products across different Amazon categories.

Key Insight:

Electronics and accessories dominate product listings, indicating high competition and demand in these segments.

Business Value:

Helps sellers identify saturated and niche markets.



Visualization 2 : Rating Distribution

Chart Type: Histogram

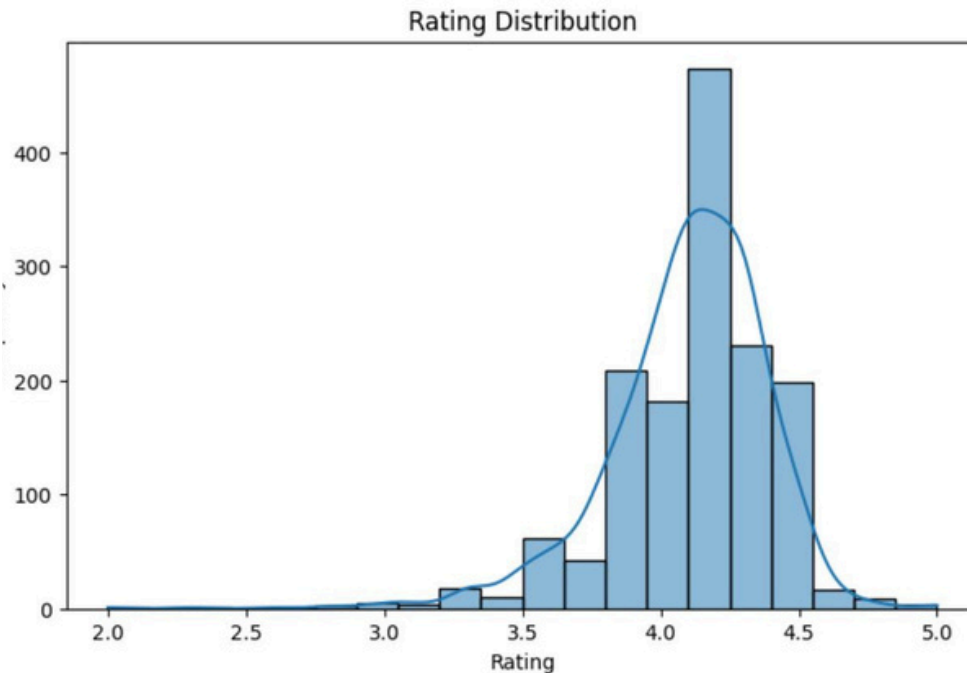
Tool: Matplotlib

What it shows:

Distribution of product ratings.

Key Insight:

Most products have ratings between 3.5 and 4.5, indicating generally positive customer satisfaction. Business Value: Helps assess overall product quality perception.



Visualization 3: Top 10 Categories With Highest Average

PriceChart Type: Bar Chart

Tool:Seaborn

What it shows:

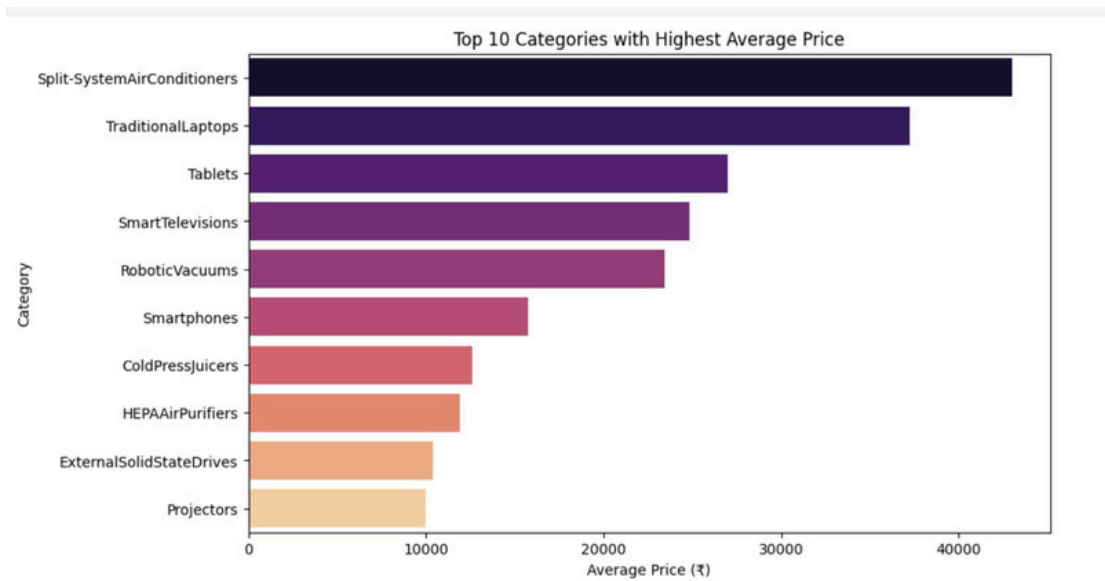
Highest Average product price for each category.

Key Insight:

Premium pricing is concentrated in electronics and large appliances, while accessories remain low-cost.

Business Value:

Useful for pricing strategy and category positioning.



Visualization 4: Top 10 Products with Highest Discounts

Chart Type: Horizontal Bar Chart

Tool: Seaborn

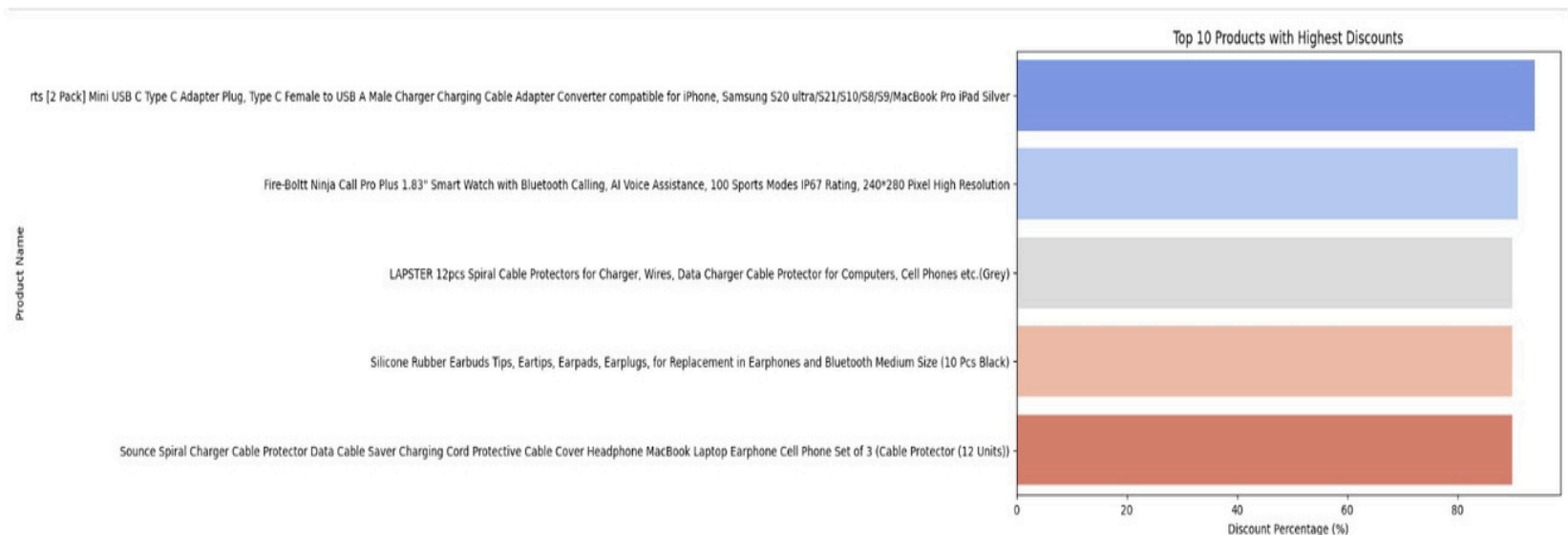
What it shows:

Products offering the highest discount percentages.

Key Insight:

High discounts are commonly used for accessories and mid-range products.

Business Value:-Supports promotional and marketing decisions.



Visualization 5: Top 10 Highest Rated Products Chart

Type:-**Horizontal Bar Chart**

Tool: Seaborn

What it shows:

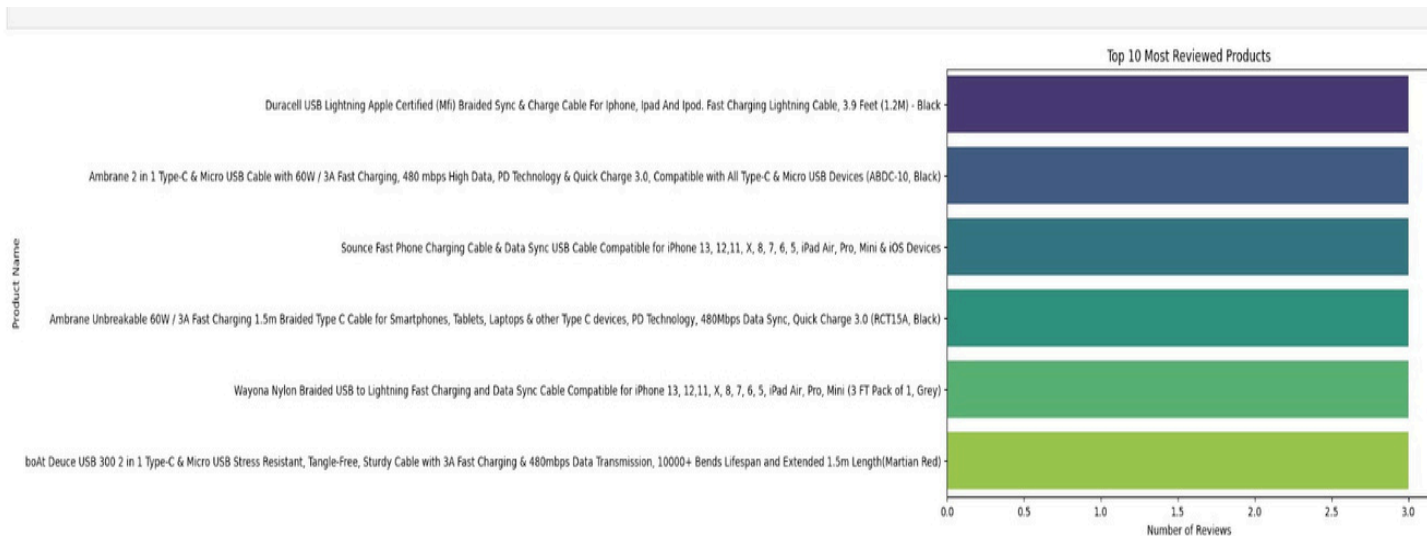
Products with the highest customer ratings.

Key Insight:

High ratings are not always linked to high prices.

Business Value:

Highlights customer-preferred products independent of price.



Visualization 6: Correlation Heatmap

Chart Type: Heatmap

Tool: Seaborn

What it shows:

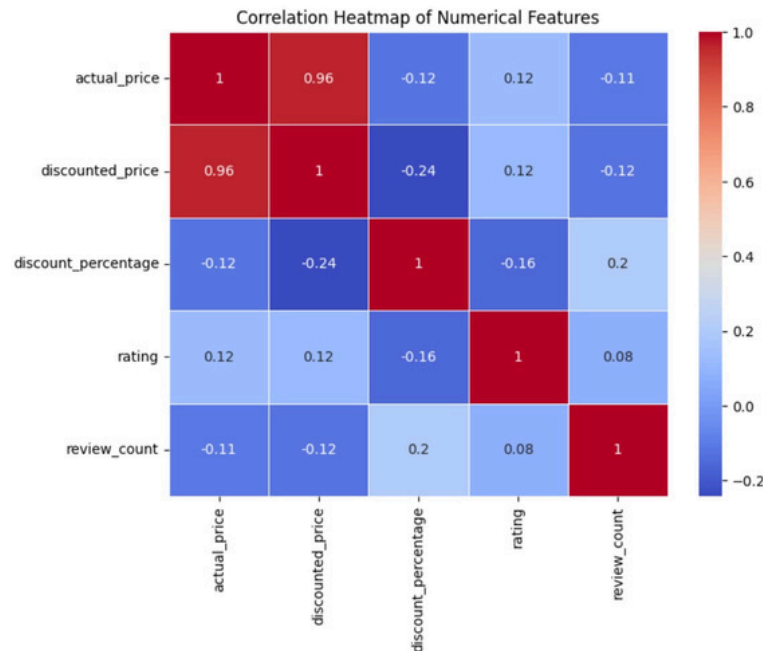
Correlation between numerical features such as price, rating, and discount.

Key Insight:

Discounts influence engagement more than ratings.

Business Value:

Supports strategic pricing and promotion planning.



Visualization 7 : Histogram / KDE Plot

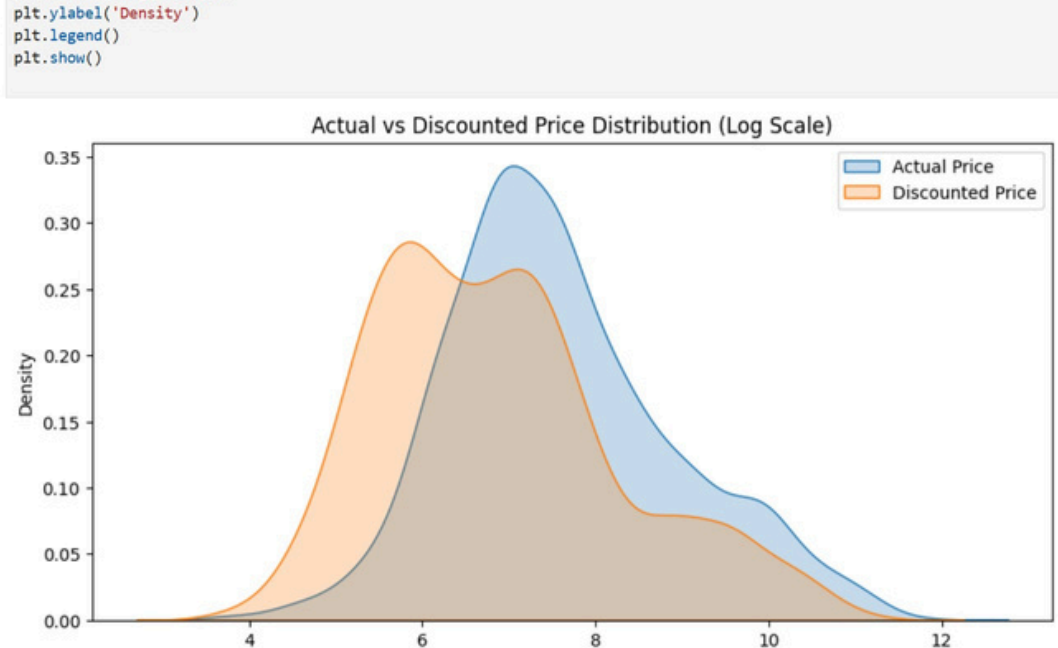
Tool: Seaborn + Matplotlib

What it shows:

Comparison of actual prices and discounted prices.

Key Insight:

Most products are clustered in lower price ranges, but discounts significantly shift purchasing attractiveness. Business Value: Shows the impact of discounting on customer perception.



- Data Story Summary (MOST IMPORTANT SLIDE)

Amazon is dominated by electronics and accessories

Discounts play a crucial role in attracting customers

Higher prices do not guarantee higher ratings Customer

satisfaction remains stable across price ranges Visualizations

reveal that value perception is more important than cost.

- Conclusion

This visualization project demonstrates how well-designed charts can:-

- Simplify complex data Highlight actionable insights Support informed business decisions The project fulfills all Task 3 requirements by using Seaborn and Matplotlib to create impactful, insight-driven visualizations.